Mehrdad Moghimi

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Summary

Machine learning researcher and final-year PhD candidate with five years of reinforcement learning research, spanning theoretical foundations and scalable algorithms for safe online and offline RL. Currently working on post-training of large language models, focusing on supervised fine-tuning and on- and off-policy knowledge distillation for mathematical reasoning, with end-to-end pipelines for generation, verification, training, and evaluation.

Research Experience

Machine Learning Research Intern, RBC Borealis, Toronto, Canada — May 2026 – Present

- Conducting research on LLM post-training via knowledge distillation, transferring reasoning capabilities from large teacher models to smaller student models using supervised fine-tuning (SFT), off-policy, and on-policy distillation objectives.
- Built an end-to-end LLM post-training pipeline using Best-of-N and Sequential Monte Carlo methods for training-data generation, followed by fine-tuning and evaluation.
- Designing and analyzing experiments on sampling efficiency, training-data selection, and teacher–student capability gaps, and investigating limitations of on-policy distillation.

Research Collaboration with **Bernardo Ávila Pires (Google DeepMind)** — Sep 2025 – Present

- Utility-Constrained Policy Optimization — *Under review*
  - Developed a practical framework for utility-constrained policy optimization using stock-augmented distributional RL.
  - Designed a safe Lagrangian actor–critic algorithm with test-time controlled safety budgets.
  - Implemented and evaluated the method on 22 Safety Gymnasium benchmarks, achieving state-of-the-art performance.

Research Assistant, York University, Toronto, ON, Canada — Sep 2021 – Present

- Theoretical Foundations of Risk-Sensitive Reinforcement Learning with Spectral Risk Measures — *ICML 2025*
  - Studied time inconsistency in risk-sensitive reinforcement learning and developed a novel theory to characterize the behavior of risk-sensitive policies.
  - Designed a time-consistent risk-sensitive RL algorithm and established convergence guarantees.
  - Validated theoretical findings through empirical evaluation on various RL environments.
- Scalable Risk-Sensitive Actor–Critic for Online and Offline RL — *Expert Systems with Applications*
  - Expanded the risk-sensitive framework with spectral risk measures to high-dimensional problems by formulating actor–critic algorithms suitable for both online and offline learning.
  - Established convergence guarantees for the policy gradient updates in the tabular setting.
  - Extended a risk-neutral JAX codebase to incorporate risk-sensitive objectives, and conducted extensive empirical evaluations demonstrating the algorithm's performance in both online and offline benchmarks.
- Joint Analysis of Time and Risk Preferences in Reinforcement Learning — *Under review*
  - Investigated the interplay between agent time preferences (via general discount functions) and risk-sensitivity to create more human-aligned agents.
  - Demonstrated that existing frameworks for hyperbolic discounting can induce time-inconsistent behavior, leading to suboptimal policies.
  - Developed a time-consistent formulation, resulting in significant performance improvements over standard hyperbolic baselines in various environments, including Atari.

Research Assistant, RiskLab, Tehran, Iran — Sep 2019 – Aug 2021

- Developed "Encoded Value-at-Risk," a novel machine learning framework utilizing Variational Auto-Encoders (VAEs) to model non-linear dependencies in high-dimensional financial data.

Research Assistant, Sharif Image Processing Lab, Tehran, Iran — Jan 2018 – Aug 2018

- Bachelor's thesis: designed a computer vision pipeline for dynamic advertisement replacement in broadcast soccer footage, using planar homography estimation to localize field banners under camera perspective.

Publications

- **M. Moghimi**, B. A. Pires, “Utility-Constrained Policy Optimization”, Under review (Paper) (Code)
- **M. Moghimi**, A. Coache, H. Ku, “Decoupling Time and Risk: Risk-Sensitive Reinforcement Learning with General Discounting”, Under review (Paper) (Code)
- **M. Moghimi**, H. Ku, “Risk-sensitive Actor-Critic with Static Spectral Risk Measures for Online and Offline Reinforcement Learning”, ***Expert Systems with Applications***, 2026 (Paper) (Code)
- **M. Moghimi**, H. Ku, “Beyond CVaR: Leveraging Static Spectral Risk Measures for Enhanced Decision-Making in Distributional Reinforcement Learning”, ***ICML 2025*** (Paper) (Code)
- H. Arian, **M. Moghimi**, E. Tabatabaei, and S. Zamani, “Encoded Value-at-Risk: A machine learning approach for portfolio risk measurement”, ***Mathematics and Computers in Simulation***, 2022 (Paper)

Education

Ph.D. in Applied Mathematics, **York University**, Toronto, Canada — Sep 2021 – Sep 2026 (Expected)
- Supervisor: Prof. Hyejin Ku | Research focus: Safe and Risk-Sensitive Reinforcement Learning

MBA in Finance, **Sharif University of Technology**, Tehran, Iran — Sep 2018 – Sep 2021
- Supervisor: Prof. Hamid Arian | Research focus: Machine Learning for Finance

BSc in Computer Science, **Sharif University of Technology**, Tehran, Iran — Sep 2014 – Sep 2018
- GPA: 18.71/20 (**Ranked 1st in the class of 2018**)

Technical Skills

- **Research Areas:** Reinforcement Learning, LLM Post-Training, Knowledge Distillation, Safe and Risk-Sensitive RL
- **Programming:** Python (proficient); R, MATLAB, Java (intermediate)
- **ML Frameworks:** PyTorch, JAX, Hugging Face Transformers, TRL, vLLM

Honours and Awards

- **York University Graduate Scholarship**, Fall 2021
- **Exceptional Talents Scholarship**, Direct admission to graduate studies (waived National Entrance Exam) due to top academic performance.